

- 8 (a)** α -particle is 2 protons and 2 neutrons; β^+ -particle is positive electron/positron
 α -particle has charge $+2e$; β^+ -particle has $+e$ charge
 α -particle has mass $4u$; β -particle has mass $(1/2000)u$
 α -particle made up of hadrons; β^+ -particle a lepton

any three

B3 [3]

(b) ${}^1_1\text{p} \rightarrow {}^1_0\text{n} + {}^0_1\beta + {}^0_0\nu$

all terms correct

M1

all numerical values correct (ignore missing values on ν)

A1 [2]

(c) (i) 1. proton: up, up, down / uud

B1

2. neutron: up, down, down / udd

B1 [2]

(ii) up quark has charge $+2/3$ (e) and down quark has charge $-1/3$ (e)
total is $+1(e)$

B1 [1]